

Nonorthogonal Cartan Schur and rational shifted-Hessian boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-084 left two analytic terms between the regular low-object theorem and a complete regular packet lower bound: the far-projected Cartan Ornstein–Uhlenbeck gradient sum and the nonlinear rational Pauli–Fierz row. This note advances both without asserting either final bound.

For Cartan CFAR, the output-orthogonality premise tested in R-082 and refuted in R-083 is removed. A weighted Cauchy–Schwarz argument gives a genuinely nonorthogonal causal Schur lemma. It spends every input weight once whenever the complete mixed Cartan atom has spatial off-diagonal decay with exponent $s > 1/2$. At the convenient production audit values

$$s = \frac{7}{12}, \quad \eta = \frac{1}{12}, \quad (1.1)$$

the exact Schur constant is $572.4472106721531\dots$ and the gap factor is $2^{-7C/6}$. This is a conditional implication, not the missing production atom theorem. The already proved first-order gain is only $3\alpha - 1 = 1/5$ at $\alpha = 2/5$. A tempting second-order value $4\alpha - 1 = 3/5$ is available only after a new cancellation theorem for the complete mixed root derivative. No cited authority proves that theorem.

For rational NEAR, the exact endpoint is expanded one order farther. Five unshifted model families have exactly the R-084 Young totals

$$\frac{7}{20}, \quad \frac{7}{10}, \quad \frac{13}{30}, \quad \frac{3}{5}, \quad \frac{23}{30}. \quad (1.2)$$

They are therefore arbitrarily form-absorbable by the existing R-063, R-068, and R-071 estimates. The expansion also exposes one signed shifted-Hessian pair. It is zero for every linear row and nonzero for the production rational row. This pair, together with its retained positive square, is the exact remaining rational NEAR target.

No production Cartan atom estimate, controlled Cartan CFAR, complete signed NEAR theorem, complete regular packet bound, overlap/revisit theorem, controlled-shell one-use estimate, $q = 10/9$ Nelson moment, measure theorem, or Sector-A closure is asserted. The tier remains T4.

2 Authorities, variables, and coefficients

The load-bearing authorities are R-050 for the production current, R-063 for the complete covariance-normal coefficient model, R-068 for centered quadratic-form duality, R-071 for smooth matrix one-forms, R-079 for complete endpoint reassembly, R-080 for the two regular low objects, R-081 for the Cartan tail and progressive boundary, R-082 for the stopped FAR and global Pauli–Fierz coordinate, R-083 for controlled polynomial CFAR, and R-084 for the root-diagonal Cartan OU identity and the complete linear-row form bound.

The production coefficients are

$$P = 4 + 10^{-12}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha_R = \frac{5}{9}. \quad (2.1)$$

In six real coordinates let P_{dbl} project onto the four doublet coordinates and put

$$r = |P_{\text{dbl}}z|^2, \quad d = |z|^2 + \epsilon_\rho, \quad g(z) = P_{\text{dbl}}z - \alpha_R \frac{r}{d} z. \quad (2.2)$$

The rational row coefficient and Gram matrix are

$$\kappa = 2\sqrt{c_1}, \quad C_R(z)y = \kappa g(z)^T y, \quad B_R(z) = \kappa^2 g(z) \otimes g(z). \quad (2.3)$$

For value heat covariance Σ define

$$\overline{B}_\Sigma = P_\Sigma B_R. \quad (2.4)$$

For the translated field and derivative put

$$U = \text{base field}, \quad a = \text{control field}, \quad G = DU, \quad c = Da, \quad Q = G \otimes G - \Gamma. \quad (2.5)$$

All endpoint identities below are finite-cutoff pathwise algebra. The form statements use the regular mutually orthogonal strict-past one-shot class of R-080–R-084.

3 The complete Cartan mixed variation

Let Φ be one production Cartan coefficient, let w contain the past and fresh value root, and let a be predictable at that root. The Cartan current increment may be written as a translated coefficient-current difference. Differentiating it in a fresh Malliavin direction v gives the single mixed second variation

$$\boxed{\mathbb{D}_j H[v] = \int_0^1 \{ D^2 \Phi(w + sa)[a, v]^T D(w + sa) + D\Phi(w + sa)[v]^T Da + D\Phi(w + sa)[a]^T Dv \} ds.} \quad (3.1)$$

The three lines are respectively the coefficient-Jacobian difference, the predictable derivative injection, and the value-innovation channel in R-084. Formula (3.1) is one object: estimating the three lines separately destroys the translated cancellation that a far-support theorem would need. Predictability gives $\mathbb{D}_j a = 0$; it does not set Da to zero.

R-084 reduces controlled Cartan CFAR to

$$\sum_{A,j} \int_0^\infty e^{-2t} \mathbb{E} \| Q_{j,C} P_t^{(j)} \mathbb{D}_j H_{A,j,A^{(j)}} \|_{\text{HS}}^2 dt \leq K_e 2^{-\theta C} \{ 1 + \mathbb{E}[X^{1/2} Y^{1/2}] \}. \quad (3.2)$$

The exact identity (3.1) sharpens what must be decomposed in (3.2), but does not prove its spatial decay.

Why second-order parilinearisation is not yet a theorem

At the production regularity $\alpha = 2/5$, the proved first-order support-refined current remainder has gain

$$3\alpha - 1 = \frac{1}{5} < \frac{1}{2}. \quad (3.3)$$

A formal order-two remainder would suggest

$$4\alpha - 1 = \frac{3}{5} > \frac{1}{2}. \quad (3.4)$$

However, the root derivative in (3.1) can remove one regularity leg. To use (3.4), one must prove that every principal term of order at most two cancels after the complete stopped-current sum, the heat compensator, both predictable drifts, and far projection are assembled. One must also prove a locally uniform translated-model norm for random H^2 controls. R-050, R-063, and R-081 prove neither assertion. Thus $3/5$ is a candidate atom exponent, not an accepted production estimate.

4 Nonorthogonal weighted causal Schur lemma

Theorem 4.1

Let $u_{j,k}$ be Hilbert-valued atoms for $k \leq j$ and set

$$f_j = \sum_{k \leq j} u_{j,k}. \quad (4.1)$$

Assume that for output shells $m \geq j + C$,

$$\mathbb{E} \|\Pi_m u_{j,k}\|^2 \leq 2^{-2s(m-k)} q_k, \quad s > \frac{1}{2}. \quad (4.2)$$

No orthogonality between the $u_{j,k}$ is assumed. For every $0 < \eta < 2s - 1$,

$$\boxed{\sum_m \sum_{j \leq m-C} 2^j \mathbb{E} \|\Pi_m f_j\|^2 \leq C_{s,\eta} 2^{-2sC} \sum_k 2^k q_k,} \quad (4.3)$$

where

$$\boxed{C_{s,\eta} = \frac{1}{(1 - 2^{-\eta})(1 - 2^{-2s})(1 - 2^{1+\eta-2s})}.} \quad (4.4)$$

Proof. Weighted Cauchy–Schwarz gives

$$\left\| \sum_{k \leq j} \Pi_m u_{j,k} \right\|^2 \leq \frac{1}{1 - 2^{-\eta}} \sum_{k \leq j} 2^{\eta(j-k)} \|\Pi_m u_{j,k}\|^2. \quad (4.5)$$

Insert (4.2), use nonnegativity to exchange the three sums, and write $j = k + r$:

$$\begin{aligned} & \sum_m \sum_{j \leq m-C} 2^j \mathbb{E} \|\Pi_m f_j\|^2 \\ & \leq \frac{1}{1 - 2^{-\eta}} \sum_k q_k \sum_{r \geq 0} 2^{k+r} 2^{\eta r} \sum_{m \geq k+r+C} 2^{-2s(m-k)} \\ & = \frac{2^{-2sC}}{(1 - 2^{-\eta})(1 - 2^{-2s})} \sum_k 2^k q_k \sum_{r \geq 0} 2^{(1+\eta-2s)r}. \end{aligned} \quad (4.6)$$

The last series converges exactly when $\eta < 2s - 1$, proving (4.3). $\square$

For fixed s , the two factors containing η are balanced at

$$\eta = \frac{2s - 1}{2}. \quad (4.7)$$

Taking (1.1) gives

$$C_{7/12, 1/12} = 572.4472106721531 \dots, \quad 2^{-2sC} = 2^{-7C/6}. \quad (4.8)$$

The executable checks four aligned worst-direction fixtures and four independent changing-sign vector fixtures against (4.3).

Sharp boundary

At $s = 1/2$, take one input u_0 with $\mathbb{E}\|\Pi_m u_0\|^2 = 2^{-m}$ and $f_j = u_0$. The truncation through shell M is

$$\sum_{m=C}^M 2^{-m} \sum_{j=0}^{m-C} 2^j = \sum_{m=C}^M \{2^{1-C} - 2^{-m}\}, \quad (4.9)$$

which grows linearly in M . Hence $s > 1/2$ is not a cosmetic constant choice for this triangular architecture.

Exact conditional Cartan contract

To apply Theorem 4.1 to (3.2), the complete mixed variation (3.1) must be decomposed into atoms $u_{A,j,k,t}$ satisfying the integrated analogue

$$\sum_A \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m u_{A,j,k,t}\|_{\text{HS}}^2 dt \leq K 2^{-2s(m-k)} q_k \quad (4.10)$$

with $s > 1/2$ and

$$\sum_k 2^k q_k \lesssim 1 + \mathbb{E}[X^{1/2} Y^{1/2}]. \quad (4.11)$$

Equations (4.10)–(4.11), with the exact normalization inherited from the R-082 stopped square, are the missing production atom theorem. Theorem 4.1 then supplies the root/input summation without output orthogonality. It does not supply (4.10).

5 Exact rational endpoint and Taylor boundary

The heat-lifted rational row functional is

$$\mathcal{W}_R(z, y) = \frac{1}{2} \bar{B}_\Sigma(z) : (y \otimes y - \Gamma). \quad (5.1)$$

Theorem 5.1 (exact endpoint)

At every smooth cutoff,

$$\begin{aligned} \Delta \mathcal{W}_R = & \frac{1}{2} [\bar{B}_\Sigma(U + a) - \bar{B}_\Sigma(U)] : Q \\ & + G^T \bar{B}_\Sigma(U + a) c + \frac{1}{2} c^T \bar{B}_\Sigma(U + a) c. \end{aligned} \quad (5.2)$$

Proof. Expand $(G + c) \otimes (G + c) - \Gamma$, subtract the base term, and group the two mixed tensors by symmetry of $\bar{B}_\Sigma$. The last term is nonnegative because heat preserves positive semidefiniteness. $\square$
Put

$$K_t = D^2 \bar{B}_\Sigma(U + ta) - D^2 \bar{B}_\Sigma(U). \quad (5.3)$$

Taylor's formula in (5.2) gives the exact seven-part split

$$\begin{aligned} \Delta \mathcal{W}_R = & \frac{1}{2} D \bar{B}(U)[a] : Q + \frac{1}{4} D^2 \bar{B}(U)[a, a] : Q \\ & + G^T \bar{B}(U)c + G^T D \bar{B}(U)[a]c + \frac{1}{2} G^T D^2 \bar{B}(U)[a, a]c \\ & + \int_0^1 (1-t) \left\{ \frac{1}{2} K_t[a, a] : Q + G^T K_t[a, a]c \right\} dt \\ & + \frac{1}{2} c^T \bar{B}(U + a)c. \end{aligned} \quad (5.4)$$

Here and below $\bar{B} = \bar{B}_\Sigma$. Define the signed shifted Hessian packet

$$\boxed{\mathcal{H}_R = \int_0^1 (1-t) \left\{ \frac{1}{2} K_t[a, a] : Q + G^T K_t[a, a]c \right\} dt.} \quad (5.5)$$

For a linear Pauli–Fierz row, $\bar{B}$ is quadratic and $K_t = 0$. Thus (5.5) is precisely the nonlinear term absent from R-084.

6 The five unshifted families are paid

The first two terms in (5.4) are the R-063/R-068 covariance-normal models; the next three are R-071 smooth one-forms with zero, one, or two undifferentiated controls. With

$$X = \|a\|_{H^2}^2, \quad Y = \|U + a\|_6^6, \quad (6.1)$$

their pathwise bounds have the following exponent ledger:

family	X exponent	Y exponent	total/slack
$D \bar{B}(U)[a] : Q$	11/40	3/40	7/20 / 13/20
$D^2 \bar{B}(U)[a, a] : Q$	11/20	3/20	7/10 / 3/10
$G^T \bar{B}(U)c$	2/5	1/30	13/30 / 17/30
$G^T D \bar{B}(U)[a]c$	2/5	1/5	3/5 / 2/5
$G^T D^2 \bar{B}(U)[a, a]c$	2/5	11/30	23/30 / 7/30

(6.2)

Every total is strictly below one. The required random moments are bounded by the worst value

$$\frac{1}{7/30} = \frac{30}{7}, \quad (6.3)$$

which the accepted model supplies.

Theorem 6.1 (five-family form absorption)

For every $\eta, \zeta > 0$, the sum $\mathcal{U}_R$ of the first five terms in (5.4) obeys

$$\boxed{\mathbb{E} \mathcal{U}_R \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta, \zeta, \epsilon_\rho, j_0}} \quad (6.4)$$

uniformly in the terminal cutoff on the regular mutually orthogonal strict-past one-shot class.

Proof. The fixed floor makes $\overline{B}$ and its derivatives smooth with the polynomial-growth bounds required by R-063 and R-071. Apply the five accepted model estimates represented in (6.2), transfer pure-control L^6 powers to Y plus Gaussian moments, and use one weighted Young allocation. The positive slacks in (6.2) give (6.4). $\square$

Combining (5.4) and (6.4), complete rational NEAR is reduced exactly to a signed lower bound for

$$\boxed{\mathcal{H}_R + \frac{1}{2}c^T \overline{B}(U + a)c.} \quad (6.5)$$

The positive square must remain coupled to $\mathcal{H}_R$; discarding it may destroy the only available square completion.

7 Why the shifted Hessian cannot be deleted

Production scalar-ray diagnostic

On a real doublet ray set

$$f_e(x) = x - \alpha_R \frac{x^3}{x^2 + e}, \quad B_R(x) = 4c_1 f_e(x)^2. \quad (7.1)$$

Direct differentiation gives

$$\boxed{B_R'''(\sqrt{e}) = \frac{6\alpha_R^2 c_1}{\sqrt{e}}.} \quad (7.2)$$

At $e = 10^{-12}$ this is

$$14062.499999996484375 \dots \neq 0. \quad (7.3)$$

Thus the rational Gram is not quadratic, K_t is genuinely present, and a five-family truncation is not an identity. Moreover, a high-frequency test of the signed Q member creates a leading cubic coefficient proportional to $B_R'''(\sqrt{e})\delta^3 N^s$. Boundedness of $D^2 B_R$ alone does not turn K_t into a control-independent multiplier against the distribution Q .

Rational-only fixed-Schur no-go

There is also an exact path for which all three linear Pauli–Fierz rows vanish. Take

$$z_0 = e_1 + e_3, \quad z_1 = C e_1, \quad y = e_3, \quad c = 0, \quad \Sigma = 0. \quad (7.4)$$

Put $q_{22} = c_1 \alpha_R^2 = 3/(320P)$ and $d_0 = 2 + e$. The baseline rational square and exact first variation are

$$F_0 = \frac{2q_{22}}{d_0^2}, \quad D_R(C) = -\frac{2q_{22}\{4C(e+1) - 5e - 2\}}{d_0^3}. \quad (7.5)$$

Consequently

$$\frac{D_R(C)}{F_0} = -\frac{4C(e+1) - 5e - 2}{e+2} \longrightarrow -\infty. \quad (7.6)$$

At $C = 2$,

$$F_0 = 0.00117187499999853515625 \dots, \quad (7.7)$$

$$D_R = -0.00351562499999560546875 \dots = -3F_0. \quad (7.8)$$

Therefore neither positivity, recombination with the vanished linear rows, nor a fixed-multiple rational-square Schur estimate proves rational NEAR. This is a method no-go, not a counterexample to a form bound: the control cost grows quadratically in C .

This fires NG-2026-07-25-A13-RATIONAL-PF-FIVE-DEGREE-AND-FIXED-SCHUR. It forbids deleting $\mathcal{H}_R$ and forbids a fixed-multiple positivity shortcut. It does not forbid a coupled shifted-Hessian form estimate.

8 The four-stage completion map

The remaining synthesis is sequential, not a collection of interchangeable hypotheses.

REG. Prove (4.10)–(4.11) for the complete mixed Cartan atom and prove the signed bound (6.5). Reassemble these with the R-080 low objects, R-083 polynomial CFAR, R-084 linear NEAR, heat traces, forest terms, and paid subtraction. Only then is the complete regular packet lower bound available.

OVERLAP. For every finite cutoff, bounded cylindrical simple predictable control, time partition, and revisit multiplicity, prove the partition-uniform full packet inequality

$$\mathbb{E} G_J(X_J + h_J(u)) + 0.45 \mathbb{E} \sum_k |I_k| \|u_k\|_2^2 \geq -C, \quad (8.1)$$

where

$$G_J(\phi) = V_J^{\text{ren}}(\phi) + 0.15 \|\phi\|_6^6. \quad (8.2)$$

Regular one-shot estimates do not imply (8.1): the class inclusion has the wrong direction and temporal packets have overlapping physical covariance ranges.

CORE. Extend (8.1) from the bounded-simple admissible core to all finite-energy progressive controls. The extension must include finite-objective truncation, auxiliary conditioning, terminal L^6 graph continuity, and lower semicontinuity. Plain L^2 predictable density is insufficient and can create infinity minus infinity.

BD. After OVERLAP and CORE, Boue–Dupuis is exact bookkeeping:

$$\inf_u \mathbb{E}\{G_J(X_J + h_J(u)) + 0.45\mathcal{E}(u)\} = -0.9 \log \mathbb{E} e^{-(10/9)G_J}. \quad (8.3)$$

The current budget arithmetic is

$$0.27 - 0.06 - 0.15 = 0.06, \quad (8.4)$$

$$\frac{1}{2(11/10)} - 0.45 = \frac{1}{220}, \quad \frac{10}{9} - \frac{11}{10} = \frac{1}{90}. \quad (8.5)$$

Thus $q = 10/9$ follows after one-use. Equation (8.3) cannot be used to prove one-use: doing so would assume the desired Nelson moment. The older $\epsilon_6 = 0.13$ allocation is superseded and is not mixed into (8.4).

9 Proof-and-failure map

route	verdict	reason or surviving target
output orthogonality	failed	R-083 smoothing witness
root orthogonality alone	failed	R-084 cumulative tree
nonorthogonal Schur	proved conditional	needs $s > 1/2$ atom
first-order Cartan tail	insufficient	$3\alpha - 1 = 1/5$
formal order-two tail	open	mixed principal cancellation absent
linear PF NEAR	closed regular	R-084
five rational families	closed regular	Theorem 6.1
delete shifted Hessian	failed	$B_R'''(\sqrt{e}) \neq 0$
fixed-square Schur	failed	$D_R/F_0 \rightarrow -\infty$
coupled rational packet	open	$\mathcal{H}_R + \frac{1}{2}c^T \bar{B}c$
regular to progressive	not implied	overlap/revisit theorem needed
BD as proof	circular	exactly equivalent after CORE

This map records both the successful reductions and the reasons the tempting shortcuts fail. A future proof should start at (4.10) and (6.5), not rerun the refuted orthogonality, five-term deletion, or fixed-Schur routes.

10 Devil’s-advocate

1. **DISMISSED: Theorem 4.1 secretly assumes orthogonality.** Equation (4.5) is weighted Cauchy–Schwarz and retains every cross term through a deterministic majorant. No probability or output orthogonality is used.
2. **VALID WITH MITIGATION: the constant 572.447... proves Cartan CFAR.** It does not. It is the exact constant in a conditional Schur implication. The production atom estimate (4.10) is explicitly unproved.
3. **UPHELD AGAINST PROMOTION: $4\alpha - 1 = 3/5$ is already a production decay theorem.** The mixed root derivative may remove a regularity leg. Complete order-two principal cancellation and a translated adapted norm theorem are missing.
4. **DISMISSED: the rational endpoint has more than the seven families in (5.4).** Taylor’s integral formula is exact. The first five, $\mathcal{H}_R$, and the final square exhaust (5.2).
5. **UPHELD AGAINST FIVE-TERM CLOSURE: bounded D^2B_R permits deleting $\mathcal{H}_R$.** The coefficient multiplies the distribution Q and depends on the shifted random field. Equations (7.2)–(7.3) show the remainder is nonzero; boundedness is not a multiplication theorem.
6. **VALID WITH MITIGATION: the rational fixture disproves every form bound.** It disproves positivity and fixed-multiple square Schur only. Its control cost grows like C^2 , so arbitrary-budget form absorption is not falsified.
7. **UPHELD AGAINST PROGRESSIVE CLOSURE: REG implies OVERLAP.** Regular one-shot controls form a smaller class. R-081 temporal factorisation preserves cost but not the disjoint physical ranges used by regular estimates.
8. **UPHELD AGAINST ONE-USE/NELSON CLOSURE.** Neither (8.1) nor its admissible-core extension is proved. Boue–Dupuis cannot provide the missing lower bound without circularity.

11 Executable verification

The primary program derives the Pauli–Fierz coefficients from the A1 manifest, evaluates the Schur constant and finite aligned fixtures, verifies the critical $s = 1/2$ divergence, differentiates the normalized rational Gram symbolically at the production floor, checks the exact endpoint and five-plus-Hessian-plus-square split, derives all Young slacks, and evaluates the rational-only no-go and synthesis budgets.

The non-importing audit uses high-precision directional differentiation, changing-sign two-coordinate Schur fixtures, independent matrix contraction, direct scalar-ray differentiation, and separately recomputed production budgets. It imports no code from the primary executable.

Run from the repository root:

```
python codes/foundations/  
a13_classii_nonorthogonal_cartan_schur_rational_hessian_  
boundary_verify.py
```

The child contracts are primary 50/50 and independent 36/36. The integrated verifier pins every authority and evidence hash, reruns both children, compares the common numerical quantities, checks the PDF and all status surfaces, and fails closed if Cartan CFAR, rational NEAR, REG, OVERLAP, CORE, one-use, Nelson, or Sector A is promoted.

Result footer

Result ID: A13-CLASSII-NONORTHOGONAL-CARTAN-SCHUR-RATIONAL-
HESSIAN-BOUNDARY

Precise statement:	A nonorthogonal weighted causal Schur lemma closes the triangular root/input sum conditionally on an $s > 1/2$ production atom estimate. The exact rational endpoint splits into five unshifted subcritical families, one signed shifted-Hessian pair, and one retained positive square. The five families are form-absorbed; the shifted-Hessian pair is the exact remaining rational target. Explicit production fixtures refute its deletion, positivity, and a fixed-multiple rational-square Schur shortcut.
Scope:	Fixed positive floor; finite cutoff; common real-even regulator; exact products. Five-family form theorem only on the regular orthogonal strict-past one-shot class. Schur result is conditional on its displayed atom hypothesis.
Dependencies:	A1; R-050; R-063; R-068; R-071; R-079--R-084.
Evidence grade:	ANALYTIC, EXACT, EXECUTED, CONDITIONAL ATOM ROUTE.
Reproduction command:	python codes/foundations/a13_classii_nonorthogonal_cartan_schur_rational_hessian_boundary_verify.py
Expected output:	Primary 50/50; independent 36/36; integrated and aggregate manifest-pinned PASS; exit zero.
Negative result:	NG-2026-07-25-A13-RATIONAL-PF-FIVE-DEGREE-AND-FIXED-SCHUR (method no-go only).
Falsification gate:	Failure of any mixed-variation, Schur, rational-split, hash, PDF, surface, or assertion contract.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No production Cartan atom or OU-gradient estimate, controlled Cartan CFAR, complete rational or signed NEAR, complete regular packet bound, progressive/revisit theorem, one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure.
Next required action:	Prove the complete mixed Cartan atom estimate (4.10)--(4.11) and the coupled signed rational bound (6.5); then REG, OVERLAP, CORE, and BD in order.